

		Option D: vary between 3.0 V and 6.0 V
25	B	Time taken $t = \frac{0.0500}{1.97 \times 10^7} = 2.538 \times 10^{-9} \text{ s}$ Electric field strength $E = \frac{V}{d} = \frac{3000}{0.100} = 3.0 \times 10^4 \text{ N C}^{-1}$ Vertical acceleration $a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.6 \times 10^{-19}) 30000}{(9.11 \times 10^{-31})} = 5.269 \times 10^{15} \text{ m s}^{-2}$ $\Delta h = u_y t + \frac{1}{2} a_y t^2$ $= 0 + \frac{1}{2} (5.269 \times 10^{15}) (2.538 \times 10^{-9})^2$ $= 1.7 \times 10^{-2} \text{ m}$
26	D	Magnetic flux linkage